Data Analytics in the Cloud MBD-EN2025 APRIL

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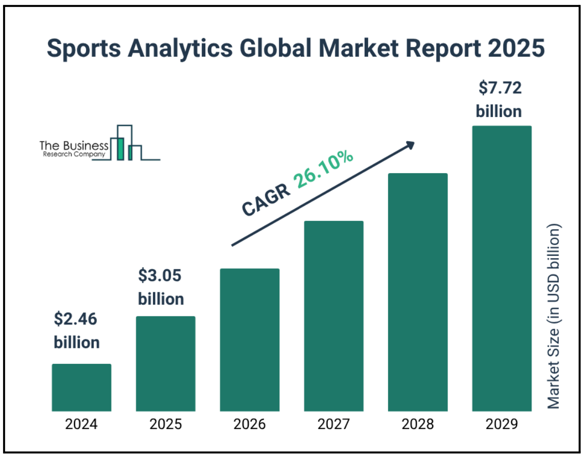
Group Work - Option A: Cloud Data Platform for Sport

Scenario

The Sports landscape has been fundamentally reshaped in recent years by global expansion and a massive influx of performance and fan metrics. Because modern supporters interact via a hybrid of live events and digital channels, organizations must now handle immense data loads instantaneously, necessitating the use of agile and scalable cloud environments. As a result, the sports analytics market size has increased exponentially over the past few years, and it is expected to grow to $7.72 billion in 2029, at a compound annual growth rate (CAGR) of 26.1%[1].



Organizations like Real Madrid[2] and Formula 1[3] are leveraging cloud platforms to centralize their data, enabling advanced analytics for player performance tracking, race-day strategy optimization, and personalized fan engagement.

Option A of the Group Work for this edition of the Data Analytics in the Cloud course will focus on a sport case. Specifically, the proposed exercise is inspired by the American National Football League (NFL) real case study[4]. Since 2017, the NFL and AWS have been at the forefront of innovation, leveraging AWS's artificial intelligence and machine learning services to shape the future of football. For this fictional scenario, we will assume that the NFL is currently deployed on-premises and is planning to migrate and modernize its data platform in the cloud.

[1] Business Research - Sport Analytics Market Report 2025
[2] Microsoft & Real Madrid: Connecting with 500 million passionate fans worldwide
[3] AWS & Formula 1: Case Study
[4] AWS & NFL: Case Study

In order to evaluate different professional and technological solutions in a comparable and objective manner, issuing a Request for Proposal (RFP) is considered a sound business practice. Providers aiming to win the project—and with it, the associated financial and reputational benefits—must respond to the RFP in detail, competing with other vendors by outlining their proposed solution, pricing, timeline, and relevant experience.

To gain hands-on experience with this type of document in the context of cloud and data analytics, each group will take on the role of a company competing in a project to migrate and modernize the data architecture of a sport business.

Materials

Alongside this group project presentation, you will find three appendices containing the necessary materials:

- **Appendix I – Request for Proposal:** Although in a simplified and more compact format, it contains the typical sections of an RFP. Read all the information carefully, including the instructions for submitting responses.
- **Appendix II – Customer Research:** Includes information that goes beyond what is typically found in an RFP, but is equally valuable for shaping your proposal. You can assume this material was gathered by your group's Business Development team through data mining and client interviews.
- **Appendix III – Sample Datasets:** Data files that can be used as a reference to identify data types and potential use cases. You are not expected to analyze or process this data. The datasets have been obtained from the "NFL 1st and Future - Analytics Competition" on Kaggle[5], where you can find more information and the terms of use. See.

Deliverables

In the Group Work activity of the course platform, each team (3-5 members) must deliver **by Feb 27, 2026 (EoD):**

1. A document with the response to the RFP, following the suggested structure.

2. An Annex to the RFP document, including:

[5] Kaggle: NFL Challenge

- References
- Generative AI acknowledgement, in accordance with the course guidelines.
- Contribution of each member to the work.

3. If the team cannot present their project in class, they must provide a ~10-15 min video with the presentation.

It is not mandatory to use AWS as the cloud service destination; other solutions can be proposed. However, students may use the Capstone Project module in the AWS Academy – IE – Data Analytics in the Cloud as a playground to explore possible architectures. Be cautious with the time and services used to avoid exhausting the system's credit.

Note that delivering a functioning solution is not a requirement of the Group Work.

Evaluation

The group project will be graded on a scale from 0 to 100, contributing proportionally to 30% of the final course grade.

Deliverable		**Points**
Project document	**Completeness & Accuracy** All points are addressed with sound responses.	20
	Clarity & Impact Responses are effective and engaging.	20
	Innovation Ideas are original and creative.	10
Video/Live presentation*		30
Individual contribution		20
TOTAL		**100**

***IMPORTANT:** All team members MUST participate in the video/live presentation.

Appendix 1. Request for Proposal

Migration and modernization of NFL Data Architecture to the Cloud

About NFL

The National Football League (NFL) is the premier professional football league in the United States, boasting a rich history of over 100 years. With 32 teams and a fanbase spanning the globe, the NFL delivers unparalleled entertainment through its games, media content, and innovative technological initiatives. The NFL events consistently rank among the most-watched programs on television, with the Super Bowl attracting over 100 million viewers annually. This immense audience reflects the NFL's pivotal role in American culture and its substantial business volume, generating billions of dollars in revenue from broadcasting rights, sponsorships, merchandise, and ticket sales.

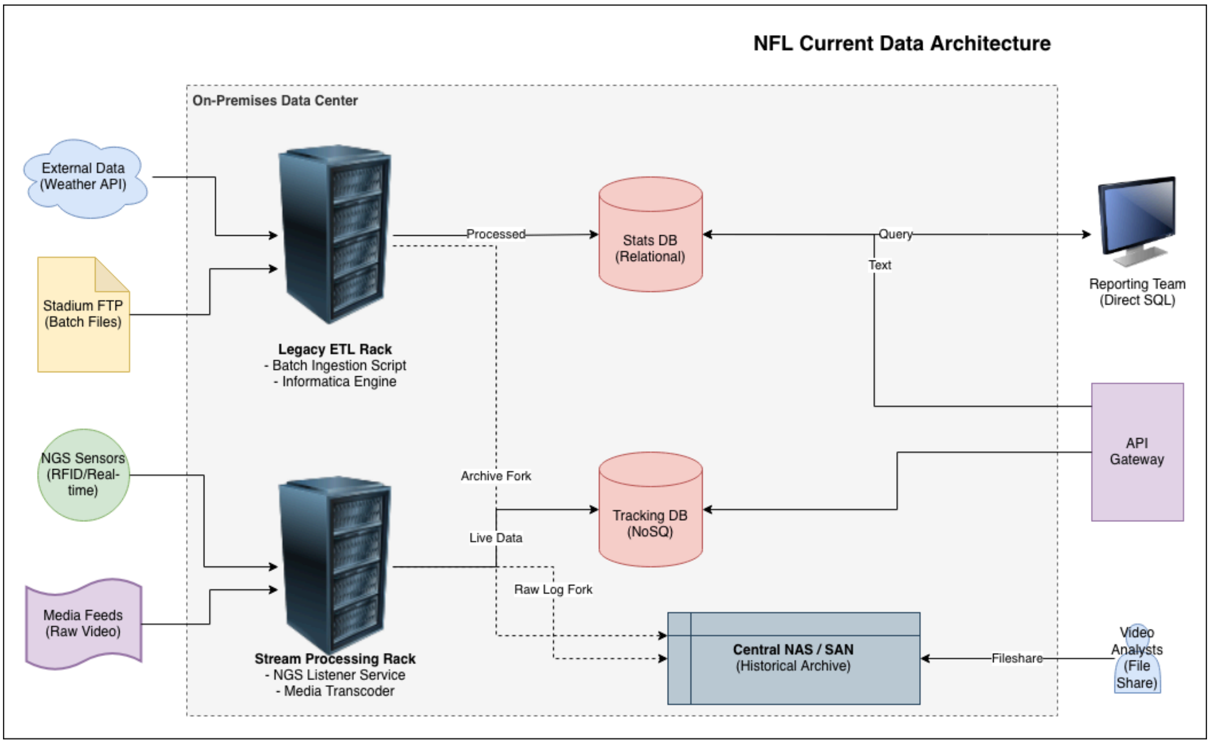
Project and Goals

The NFL maintains a substantial repository of data on-premises, encompassing 10 petabytes of media, images, and other files, representing over 30 years of games. Their operational requirements focus on frequent access to data from the past year, with data aged between one and five years accessed occasionally for monthly reporting purposes. Data older than five years is rarely utilized but is preserved for archival needs.

In addition to media files, the NFL stores relational data on players and games in an on-premises database, complemented by key-value records of player tracking information. During games, the NFL employs its Next Gen Stats system to capture real-time player tracking data, as detailed in Appendix 3.A. This data includes player-specific information such as position, direction, distance traveled, speed, orientation, and various in-game events. These data streams are processed on-premises to generate real-time visualizations, such as field heat maps and statistical insights, which are shared with broadcasters.

The NFL also integrates data from online sources, including weather services, although these inputs do not require real-time processing. After games, the league combines player tracking data with additional datasets to generate comprehensive game reports. This process involves integrating data from file-based storage, relational and non-relational databases, and external sources. Appendix 3.B provides a detailed description of the Play dataset, which combines information from weather services and the league's databases on stadiums, games, and plays to support report generation.

The following diagram illustrates NFL's current (as-is) architecture:



Source: Generated with Draw.io

Looking to the future, the NFL seeks to leverage its data assets for advanced analytics and machine learning. The organization aims to predict critical incidents, such as player injuries, by analyzing historical injury data in conjunction with player tracking information (See Appendix 3.C). This initiative focuses on assessing the predictive potential of various factors, including plays, weather conditions, and other contextual variables, to enhance player safety and operational insights. Additionally, the NFL seeks to improve customer access to their data by facilitating natural language processing (NLP) through web-based chatbots, as well as the integration of external AI agents for automated queries and actions.

Scope and Requirements

The project involves migrating and modernizing the NFL's existing on-premises data infrastructure to a secure, high-performing cloud-based architecture. The target architecture must not be a mere replica of the on-premises data platform, but introduce improvements that reflect on the business value of NFL's data. The main project requirements are described below.

R1. Timeframe and Data Synchronization: The migration must be completed within 12 months from project initiation. During this period, data consistency and real-time synchronization must be maintained to ensure that all data in the target cloud architecture is up-to-date throughout the transition.

R2. Environment Setup: Three separate environments must be provisioned and configured: Development (Dev) for testing and experimentation, Pre-Production (Pre-Prod) for staging and validation, and Production (Prod) for live operations. Anonymized samples of production data should be utilized for the development and pre-production environments to maintain realism while ensuring compliance with privacy standards.

R3. Automation: Key processes must be automated, including infrastructure creation and scaling, data lifecycle management (ingestion, processing, storage, and archiving), and end-to-end data processing workflows. Infrastructure-as-Code (IaC) tools should be leveraged for consistency and repeatability in these processes.

R4. Security and High Availability: Advanced security measures must be implemented to protect sensitive data, including encryption, role-based access control, and threat detection. High availability must be ensured with a target uptime of at least 99.9%, supported by redundant systems and failover mechanisms.

R5. Cost Optimization: The project should achieve measurable cost reductions compared to the existing on-premises platform. Resource allocation must be optimized to minimize operational expenses without compromising performance or security.

R6. Data Quality: Data quality monitoring and validation protocols should be established to ensure accuracy, consistency, and completeness. Automated checks and real-time alerts must be implemented to promptly identify and resolve data anomalies.

R7. Sustainability: Environmentally sustainable practices should be adopted in cloud operations by leveraging energy-efficient services and regions. Metrics should be provided to demonstrate improvements in sustainability compared to the on-premises infrastructure.

Structure for Proposal Responses

Proposals must adhere to the following structure to ensure clarity, consistency, and ease of evaluation. The entire response should not exceed 10 pages:

1. **Executive Summary:** The executive summary should present a concise overview of the proposal, highlighting the key business value it offers to the NFL. This section should emphasize the primary benefits, including how the proposal aligns with the NFL's goals and objectives, and provide a compelling rationale for its adoption.

2. **High-Level Architecture:** Describe the overarching design of the proposed data platform in the cloud. A clear and professionally designed diagram must illustrate the main components, showcasing their roles and interactions. The narrative should explain how these components work together to address the NFL's requirements and enable efficient data processing, analysis, and management.

3. **Low-Level Architecture**: This section should delve into the detailed technical architecture, specifying the services and technologies to be used, such as AWS or alternative cloud platforms. A comprehensive diagram should depict these services and their integrations, supported by a justification of their selection based on performance, scalability, cost-efficiency, and alignment with the NFL's needs.

4. **Project Description:** Explain how the proposed solution meets the scope and requirements outlined in the RFP. It should articulate the approach, processes, and methodologies employed to achieve the stated objectives and provide a detailed explanation of the value delivered by the solution.

5. **Examples of Data Processing, Visualization, and Predictions:** This section should include specific examples demonstrating how the proposed architecture and services address the NFL's data sources and use cases. These examples should cover data ingestion, processing workflows, visualization dashboards, predictive analytics outputs, NLP support and AI agent integrations.

6. **Future Improvements:** Propose potential enhancements to the initial proposal, identifying areas for growth and innovation. It should detail the anticipated benefits of these improvements and explain how they can sustain the platform's long-term value and adaptability to the NFL's evolving needs.

Appendix 2. Customer Research

Business research on NFL

Competitive Analysis

The professional sports market is currently experiencing intense competition, driven by rapid shifts in fan engagement, the rise of digital-native streaming platforms, and increasing pressure to maximize athlete longevity and performance. To stay ahead, organizations are leveraging data to enhance ticket and merchandise sales through personalization, optimize roster management through predictive performance analytics, and reduce operational overhead for stadium logistics. The NFL has reported solid annual revenues of $19 billion; however, its recent growth rate of 4.5% in the 18–24 demographic falls short of the broader sports entertainment market average. This gap is particularly evident in digital channels and international markets, where the league's engagement and direct-to-consumer sales growth remain underwhelming, with NFL+ subscriptions increasing only 1.8% year-over-year.

Key Decision Makers

Sarah Jenkins (she/her) – Chief Operations Officer (COO): With 20 years at the league office and a deep background in game-day logistics, Sarah is a pragmatic leader who prioritizes league integrity and on-field results over technological trends. She's open to disruptive change—but only if the business case is solid and the solution is end-to-end. Sarah demands that any modernization proposal be comprehensive, controlled-risk, and demonstrably capable of improving player safety, game-day efficiency, and global competitiveness before she gives her final approval.

Marcus Thorne (he/him) – Chief Technology Officer (CTO): Marcus has been with the NFL for 10 years and is the architect behind the current on-premises data centers. He's skeptical of total cloud migrations, viewing them as potentially costly and a threat to the league's long-standing operational control over sensitive game data. He champions hybrid models and is highly concerned about vendor lock-in. While not completely against change, he wants guarantees that modernization won't sacrifice the ultra-low latency required for live-broadcast stats or the total control he currently maintains.

Robert Vance (he/him) – Chief Financial Officer (CFO): A long-term executive with 30 years at the league, Robert has traditionally favored stable, long-term capital expenditure (CapEx) in physical infrastructure and signed the current server colocation deals. However, recent financial analysis has shown that the league is spending twice as much on data maintenance as other leagues like NHL or NBA, while achieving only half the analytical output. Robert is increasingly concerned with operational inefficiency and is open to modernization—if it can show a clear path to cost reduction, better ROI, and sustainable long-term value.

Elena Rodriguez (she/her) – Chief Information Security Officer (CISO): Elena is new to the NFL and came in with a mandate to fix weak points in data privacy and compliance. With a strong background in biometric data regulation and cybersecurity, she is laser-focused on ensuring that any future data architecture meets the highest standards of security for sensitive player medical records and legal compliance for sports betting data. After a high-profile data leak led to the previous CISO's dismissal, Elena views robust security as non-negotiable for any modernization effort.

Jordan Taylor (they/them) – Head of Analytics & AI: Jordan recently joined the NFL from a leading Silicon Valley tech firm, bringing strong experience with modern cloud analytics and real-time machine learning. They see the current on-prem architecture as outdated and overly rigid, with failing batch processes leaving Next Gen Stats outdated or incomplete, and limited access to siloed team databases. Simply "lifting and shifting" what they have to the cloud won't solve the problem. They need a modern, robust, and reliable data architecture to power the next generation of AI-driven fan experiences.

Appendix 3. Sample Datasets

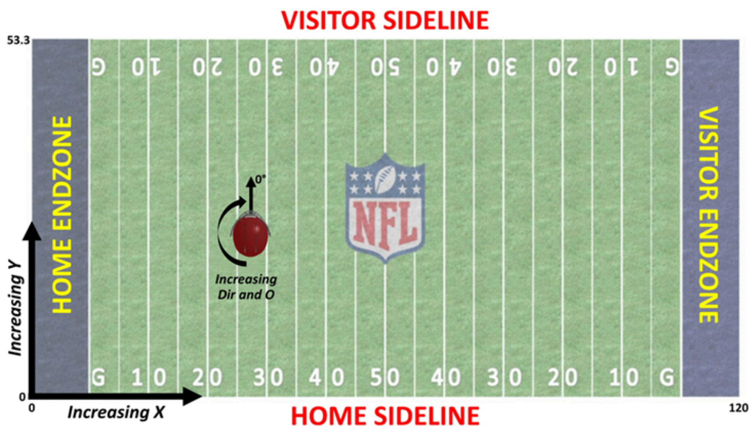
The NFL data and the example files attached to the activity on the course platform are detailed below.

A. Player Track Data

The player track file in .csv format includes a sample of player position, direction, and orientation data for player 43540 during the entire course of the play collected using the Next Gen Stats (NGS) system. This data is indexed by PlayKey (which includes information about the player and game), with the time variable providing a temporal index within an individual play.

File: **PlayerTrackData_43540.csv**

Field	Format	Description	Additional Details
PlayKey	character string		
time	float	time in seconds (float) since the start of the NGS track for the play	this is the time index for the player track
event	character string	play details (character string) as a function of time during the play (huddle break, snap, etc.)	
x	numeric list	player position (float) along the long axis of the field (yards) over time index	0 - 120 yards
y	numeric list	player position (float) along the short axis of the field (yards) over the time index	0 - 53.3 yards
dis	numeric list	distance traveled (float) from prior time point over the time index	Distance (yards)
s	numeric list	estimated speed (float) at that particular point in time over the time index	yards per second
o	numeric	Orientation (float) - angle that the player is facing (deg)	0 - 360 degrees
dir	numeric	Direction (float) - angle of player motion (deg)	0 - 360 degrees



B. Play List Data

The play file contains information about each player-play in the dataset, to include the player's assigned roster position, stadium type, field type, weather, play type, position for the play, and position group

File: **PlayList.csv**

Field	Format	Description
PlayerKey	XXXX	uniquely identifies a player with a five-digit numerical key
***GameID**	PlayerKey-X	uniquely identifies a player-game (this index is not strictly in temporal order) – see note below
PlayKey	PlayerKey-GameID-X	uniquely identifies a player's plays within a game (in sequential order within a game)
RosterPosition	character string	provides the player's roster position (i.e. Running Back)
*PlayerDay	integer	an integer sequence that reflects the timeline of a player's participation in games; use this field to sequence player participation
*PlayerGame	integer	uniquely identifies a player's games; matches the last integer of the GameID (not strictly in temporal order of game occurrence)
StadiumType	character string	a free text description of the type of stadium (open, closed dome, etc.)
FieldType	character string	a categorical description of the field type (Natural or Synthetic)
Temperature	float	on-field temperature at the start of the game (not always available - for closed dome/indoor stadiums this field may not be relevant as the temperature and weather are controlled)
Weather	character string	a free text description of the weather at the stadium (for closed dome/indoor stadiums this field may not be relevant as the temperature and weather are controlled)
PlayType	character string	categorical description of play type (pass, run, kickoff, etc.)
PlayerGamePlay		an ordered index (integer) denoting the running count of plays the player has participated in during the game
Position	character string	a categorical variable denoting the player's position for the play (RB, QB, DE, etc.) – may not be the same as the roster position.
PositionGroup	character string	a categorical variable denoting the player's position group for the position held during the play

C. Injury Data

Contains the injuries sustained by the player according to the match and the play, identified by PlayerKey, GameID, and PlayKey, respectively. Additionally, it specifies the affected body part, the type of surface (natural or synthetic) of the match, as well as the severity classified by days lost due to the injury. Note that there is not a PlayKey available for every injury. This indicates that the game in which the injury occurred is known, but the specific play in which the injury occurred was not noted at the time of injury.

File: **InjuryRecord.csv**

Field	Format	Description
PlayerKey	XXXX	Uniquely identifies a player with a five-digit numerical key
GameID	PlayerKey-X	Uniquely identifies a player's games (not strictly in temporal order)
PlayKey	PlayerKey-GameID-X	Uniquely identifies a player's plays within a game (in sequential order)
BodyPart	character string	Identifies the injured body part (Knee, Ankle, Foot, etc.)
Surface	character string	Identifies the playing surface at time of injury (Natural or Synthetic)
DM_M1	1 or 0	One-Hot Encoding indicating 1 or more days missed due to injury
DM_M7	1 or 0	One-Hot Encoding indicating 7 or more days missed due to injury
DM_M28	1 or 0	One-Hot Encoding indicating 28 or more days missed due to injury
DM_M42	1 or 0	One-Hot Encoding indicating 42 or more days missed due to injury